

Primăria

Municipal workflows with deterministic rules and tenant isolation

Project overview · Offline portfolio work · reviewed 14 September 2026

Primăria · tenant and export controls

Code-derived implementation map · not a product screenshot

Tenant context → **application** → **PostgreSQL**

Forced row-level security across 34 tenant tables

Deterministic tax rules

Buildings · land · vehicles

XML exporter modules

F3001 · F3002 · F3003 · F3101

318 offline tests + 9 real-PostgreSQL tests

Both in CI · database checks use a restricted role

Offline portfolio implementation · no export-acceptance certification

Problem

Municipal administration combines resident records, tax calculations and reporting. I modelled those responsibilities as tenant-scoped workflows for citizens and staff.

Engineering decision

Keep authoritative calculations deterministic and separate from AI assistance. Enforce tenant context in the application and database, and compare XML exports against golden-file fixtures.

My contribution

Founder and sole engineer: Romanian municipal workflow concept, Next.js/TypeScript implementation, PostgreSQL data modelling and tests.

Visual evidence

Code-derived map of tenant isolation and XML export controls; not a product screenshot or certification of export acceptance.

Engineering evidence

Dated verification, implementation scope and remaining work

Implementation

Next.js, TypeScript, React, PostgreSQL and Prisma form the primary stack. Checked-in controls include forced row-level security across 34 tenant tables and AES-256-GCM encryption for the Contribuabil.cnp field.

Tax modules cover buildings, land and vehicles. The PatrimVen exporter covers F3001–F3003 and F3101. Citizen/staff interfaces, audit records and Romanian, Hungarian and English message catalogues are present.

AI boundary

The cited fiscal assistant includes prompt-injection defence and PII redaction, routing model calls through the separate AI Gateway project. AI Gateway remains offline; it is not offered as a hosted service.

The erasure-request flow records requests and cleans up notifications. Approval, anonymisation and deactivation are deferred. These boundaries matter when assessing the privacy workflow.

Evidence to inspect

Public source: <https://github.com/Bogdan0708/PrimatIA> — 318 offline tests plus 9 real-PostgreSQL isolation tests (327 total), both run in CI. The database job uses a restricted role without superuser or RLS-bypass privileges. Golden-file exporter tests and tenant-isolation checks provide specific engineering evidence.

The diagram follows checked-in tenant controls and exporter modules. The unmodified historical UI capture, with empty export history, is available at <https://bogdan0708.github.io/assets/projects/primaria-export.png>. Neither image demonstrates a new export or production deployment.

Status and next validation

Offline portfolio/pilot-stage implementation. There is no municipal production rollout, independent security/compliance certification or ANAF acceptance claimed. UI wording describing export formats is not independent legal assurance.

Independent legal/export-format validation, deployment hardening and pilot evidence are required before production, adoption or measured-impact claims.